

Sparse Bayesian Learning for End-to-End EEG Decoding

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Abstract—Decoding brain activity from non-invasive electroencephalography (EEG) is crucial for brain-computer interfaces (BCIs) and the study of brain disorders. Notably, end-to-end EEG decoding has gained widespread popularity in recent years owing to the remarkable advances in deep learning research. However, many EEG studies suffer from limited sample sizes, making it difficult for existing deep learning models to effectively generalize to highly noisy EEG data. To address this fundamental limitation, this paper proposes a novel end-to-end EEG decoding algorithm that utilizes a low-rank weight matrix to encode both spatio-temporal filters and the classifier, all optimized under a principled sparse Bayesian learning (SBL) framework. Importantly, this SBL framework also enables us to learn hyperparameters that optimally penalize the model in a Bayesian fashion. The proposed decoding algorithm is systematically benchmarked on five motor imagery BCI EEG datasets ($N = 192$) and an emotion recognition EEG dataset ($N = 45$), in comparison with several contemporary algorithms, including end-to-end deep-learning-based EEG decoding algorithms. The classification results demonstrate that our algorithm significantly outperforms the competing algorithms while yielding neurophysiologically meaningful spatio-temporal patterns. Our algorithm therefore advances the state-of-the-art by providing a novel EEG-tailored machine learning tool for decoding brain activity.

Index Terms—Electroencephalography (EEG), brain-computer interface (BCI), emotion recognition, decoding, spatio-temporal filtering, sparse Bayesian learning.

I. INTRODUCTION

ELECTROENCEPHALOGRAPHY (EEG) is the most widely used technique for recording human electrical brain activities, with broad applications in neural engineering, neuroscience, as well as clinical settings owing to its low cost, portability, ease of setups, non-invasiveness, and the ability to capture brain signals with high temporal resolution [1]. In particular, recent years have witnessed the successful use of EEG in brain-computer interfaces (BCIs), emotion recognition, and precision psychiatry [2], [3], [4]. Across these research areas, it is crucial to accurately map the high-dimensional spatio-temporal EEG data to a target variable (e.g., motor intention in BCIs, emotion category in emotion recognition, or diagnosis/prognosis for psychiatric disorders), herein referred to as EEG decoding.

Decoding complex spatio-temporal patterns of brain activity is critical, but challenging for both invasive and non-invasive recordings. Unlike direct brain recordings, EEG also suffers from volume conduction, wherein brain activity spatially spreads across the scalp surface due to poor head tissue conductivities. Additionally, EEG has a low signal-to-noise ratio (SNR) due to a variety of physiological and non-physiological sources of interference, which are also distributed across channels at different spatial locations. Furthermore, information relevant to decoding may reside in distinct frequency bands of the EEG signals for different individuals, which may not perfectly align with the frequency range of canonical EEG rhythms. These unique spatio-spectral characteristics of EEG require the careful design of spatial and temporal filters for feature extraction, which renders EEG decoding a highly challenging problem.

Notably, while volume conduction reduces the spatial resolution of EEG, it nonetheless provides an opportunity to develop linear spatial filters for extracting brain activities from multichannel EEG. This is because volume conduction amounts to a spatial linear mixing process according to principles of physics. An assorted supervised algorithms were developed or applied to leverage information about the decoding tasks when optimizing the spatial filters. For instance, canonical correlation analysis (CCA) [5], maximum contrast combination (MCC) [6], minimum energy combination (MEC) [6] were employed to improve the signal-to-noise ratio (SNR) of the steady-state visual evoked potentials (SSVEPs) in SSVEP-based BCIs; Similarly,

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the xDAWN algorithm attempted to enhance the SNR of the event-related potentials (ERPs) in ERP-based BCIs [7]. For the spectral power feature, common spatial patterns (CSP) is a popular spatial filtering algorithm that maximizes the variance ratio between two classes, achieving tremendous success in motor imagery (MI) BCIs [8], [9].

Spatial and temporal filtering in traditional EEG decoding methods are performed at separate stages, typically with temporal filtering followed by spatial filtering. However, this sequential approach may not yield optimal decoding performance. To address this limitation, several extensions to the CSP algorithm have been proposed that optimize both the spatio and temporal filters simultaneously. For instance, CSSP extends it by using the first-order finite-impulse response (FIR) filter to incorporate spectral analysis, enabling the extraction of both spatial and spectral patterns from the data [10]. CSSSP improves upon CSSP by removing the restriction on the FIR filter order, thus allowing more flexibility in the filter design. Moreover, an l_1 -norm penalty term is added to the objective function to sparsify the temporal filter and prevent overfitting [11]. RSTFC provides a unified variance ratio maximization framework to optimize both spatial and temporal filters. It uses an l_2 -regularized algorithm for supervised spatio-temporal filtering [12]. FBCSP filters EEG data into sub-band components and performs CSP on each component to extract features [13]. On the other hand, SPECCSP and ISSPL iteratively optimize spatial and spectral filters, respectively, even though they use distinct objective functions at each stage [14], [15]. However, in existing CSP-based algorithms, the spatio-temporal filters and the classifier are optimized separately. Moreover, in conventional EEG decoding pipelines, classifiers were built following feature extraction to relate the features to the target variable. Representative classifiers for EEG decoding include support vector machine (SVM), Fisher Linear discriminant analysis (FLDA), logistic regression, K-nearest neighbors (K-NN), Naive Bayes, and neural networks such as the convolutional neural network (CNN), deep belief network (DBN) and discriminative graph regularized extreme learning machine (GELM) [3], [16]. Despite the utility of such a divide-and-conquer approach, decoding performance may not be optimal since feature extraction and classification are performed in separate stages with distinct cost functions [4], [17], [18].

In recent years, end-to-end EEG decoding has gained enormous attention due to its appeal of being able to relate the EEG data to the target variable in a direct manner. The ability of automatically learning intricate features from a large amount of data makes neural networks often the preferred choice for end-to-end EEG decoding. Specifically, Li et al. studied different deep structures based on the restricted Boltzmann machine (RBM) to learn the spatial and temporal patterns from single-trial ERP signals [19]. Additionally, CNNs achieved success in decoding motor intentions [20], [21], [22]. In particular, Schirrmeister et al. showed that the shallow CNN (sCNN) and deep CNN (dCNN) could achieve comparable performance as FBCSP for EEG decoding in MI BCIs [21]. Lawhern et al. introduced a compact CNN, named EEGNet, where depthwise and separable convolutions were employed to construct an EEG-specific

model [22]; it was reported that EEGNet is robust enough to learn a wide variety of interpretable features over a range of BCI tasks. EEG-inception is a CNN for ERP detection, which includes the use of inception modules, residual connections, and a design that introduces the symmetry of the brain through the mid-sagittal plane into the network architecture [23]. Similarly, EEGSym is another model that includes mid-sagittal symmetry and has shown promising results in EEG classification tasks [24]. However, without a principled approach to constrain the model complexity, these algorithms may overfit the data considering the large number of unknown parameters in the networks. Nonetheless, the sample size in most current EEG decoding studies is too limited to prevent generic neural networks from overfitting the data, in addition to the difficulty of interpreting the activations in the intermediate layers. Restricting the number of layers may alleviate the overfitting issue, yet determining network structures suited for EEG may not be straightforward. Moreover, gradient-based algorithms typically employed for parameter optimization in neural networks may not be able to find good solutions due to the existence of multiple local minima.

Alternatively, several algorithms have been proposed to perform end-to-end EEG decoding by exploiting the underlying linear algebraic structure of feature extraction and pattern recognition, as demonstrated in previous works [4], [17]. Convex optimization algorithms have been developed for simultaneous spatial filtering and classification, where the predicted target variable can be represented as a linear transform of the EEG spatial covariance matrix in the case of spectral power features. To guard against overfitting, a nuclear-norm regularization term has been added to the cost function to penalize the number of spatial filters. Since the optimization problem is convex, efficient algorithms can be developed to find the global optimum solution. However, the nuclear norm is only a convex surrogate for the matrix rank and may not always optimally penalize the number of spatial filters. Additionally, these algorithms rely on predetermined canonical EEG frequency bands, which may not be the most effective choice for decoding.

To address the aforementioned issues with the existing EEG decoding algorithms, this paper proposes a new algorithm, termed Sparse Bayesian Learning for End-to-end Spatio-Temporal-filtering-based single-trial EEG classification (SBLEST), for end-to-end EEG decoding. SBLEST leverages the advances in *sparse Bayesian learning* (SBL) over the past two decades [25], [26], [27], [28], particularly when it was recently employed to solve rank minimization problems [28]. The novel contributions of our algorithm are summarized below:

- Unlike existing EEG decoding algorithms, spatio-temporal filtering-based feature extraction and classification are all integrated in a linear matrix regression model through proper reparameterization of the spatio-temporal filters and the classifier. With this framing, the number of spatio-temporal filters can be conveniently controlled by penalizing the rank of the weight matrix in the linear matrix regression model.
- Model estimation is performed via SBL, which automatically encapsulates a notion of Occam's razor. As such, the regularization parameter is determined in conjunction

with model parameters, thus obviating the need of out-of-sample model selection procedures, e.g., cross-validation, which are often cumbersome and do not make efficient use of the data [29].

- A computationally efficient algorithm is developed to solve the optimization problem for model estimation. Despite the rank-minimization problem being non-convex, it has been shown that SBL is more advantageous than convex algorithms with the nuclear norm regularization [28]. Our experimental results demonstrate that this desirable theoretical property also translates to better decoding performance on real EEG datasets.

The remainder of this paper is organized as follows: Section II presents the mathematical details of SBLEST. We evaluate the classification performance of SBLEST on five motor imagery BCI datasets and one emotion recognition dataset. The performance of SBLEST is compared to that of a number of extant algorithms, including CSP-based algorithms, state-of-the-art deep learning algorithms, and convex optimization algorithms. The details of these datasets, analysis pipelines of SBLEST and performance comparisons are provided in Section III. Several important properties of SBLEST along with its limitations and future work are discussed in Section IV. Finally, Section V summarizes the paper.

II. METHOD

A. Problem Setup

Let $\mathbf{X} \in \mathbb{R}^{C \times T}$ denote a single-trial EEG signal:

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_C]^\top, \quad (1)$$

where C and T denote the number of recorded channels and the number of sampled time points, respectively; $\mathbf{x}_c \in \mathbb{R}^{T \times 1}$ is the c -th channel of the EEG signal; $\top$ denotes the transpose operator. The goal of EEG decoding is to determine an optimal mapping f from $\mathbf{X}$ to the desired target variable $y \in \mathbb{R}$:

$$y = f(\mathbf{X}) + \epsilon, \quad (2)$$

where ϵ is the additive noise. In this paper, we focus on addressing EEG decoding problems with a linear regression approach, even though our algorithm is evaluated for its performance in binary classification, where $y \in \{-1, 1\}$. An analogous prior example in the field of machine learning is least-squares classification [30].

We consider EEG decoding problems based on the spectral power feature, which is employed to quantify the strength of rhythmic brain activities that are of primary interests in many decoding applications. The spectral power feature has been extensively used for EEG decoding thanks to its low computational cost, high test-retest reliability, and well-grounded neurophysiology as documented in the literature [3], [8], [31]. In particular, in MI BCIs, the modulation of EEG rhythms, known as the event-related desynchronization/synchronization (ERD/ERS), can be well captured by changes in the spectral power [8]; In prior EEG emotion recognition studies, the differential entropy (DE), which is equivalent to the logarithmic spectral power, was used to characterize the patterns of the EEG

rhythms corresponding to different emotions [3]; In precision psychiatry, the spectral power of the pretreatment resting EEG was employed to successfully predict treatment outcome for antidepressants [4].

The spectral power features can be computed on $\mathbf{X}$ itself, i.e., in the sensor space, which is not ideal due to volume conduction, or on a representation of $\mathbf{X}$ in the latent space via spatio-temporal filtering, which is the approach taken in this paper. The spectral power features of the latent representations are then mapped to y via a pattern classifier.

We next show how a spatial filter and corresponding channel-specific temporal filters can be reparameterized by a single weight vector, and then illustrate how the spatio-temporal filters and the classifier are encoded by a low-rank weight matrix, which can be optimized under the SBL framework.

B. EEG Spatio-Temporal Filtering as a Linear Mapping

To allow for flexibility in shaping spectral bands for distinct channels, we implement channel-wise FIR temporal filtering with order K . Specifically, the c -th channel of the EEG signal $\mathbf{x}_c$ is temporally filtered to yield $\mathbf{q}_c \in \mathbb{R}^{T \times 1}$:

$$\begin{aligned} \mathbf{q}_c(t) = & a_{c,1}\mathbf{x}_c(t) + a_{c,2}\mathbf{x}_c(t - \tau) + \\ & \dots + a_{c,K}\mathbf{x}_c(t - (K - 1)\tau), \end{aligned} \quad (3)$$

where $t = 1, 2, \dots, T$, and $\mathbf{a}_c = [a_{c,1}, a_{c,2}, \dots, a_{c,K}]^\top \in \mathbb{R}^{K \times 1}$ ($c \in \{1, 2, \dots, C\}$) is the temporal filter for the c -th channel. τ is the delay time between adjacent input samples of each temporal filter; In particular, no temporal filtering is performed when $\tau = 0$.

Let $\mathbf{s} = [s_1, s_2, \dots, s_C]^\top \in \mathbb{R}^{C \times 1}$ denote the spatial filter. Performing spatial filtering on the temporally filtered EEG signal yields the spatio-temporally filtered signal (also known as the *latent signal*) $\mathbf{z} \in \mathbb{R}^{T \times 1}$:

$$\mathbf{z} = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_C] \cdot \mathbf{s}. \quad (4)$$

In light of our earlier work [12], performing spatio-temporal filtering on the EEG signal $\mathbf{X}$ can be equivalently represented as the following more compact linear operation:

$$\mathbf{z} = \tilde{\mathbf{X}}^\top \mathbf{v}, \quad (5)$$

where $\mathbf{v} \in \mathbb{R}^{KC \times 1}$ denotes the spatio-temporal filter encoding both the spatial and temporal filters:

$$\begin{aligned} \mathbf{v} = & \text{vec}[[s_1\mathbf{a}_1, s_2\mathbf{a}_2, \dots, s_C\mathbf{a}_C]^\top] \\ = & [s_1a_{1,1}, s_2a_{2,1}, \dots, s_Ca_{C,1}, \dots, \\ & s_1a_{1,K}, s_2a_{2,K}, \dots, s_Ca_{C,K}]^\top \\ = & [v_{1,1}, v_{2,1}, \dots, v_{C,1}, \dots, v_{1,K}, v_{2,K}, \dots, v_{C,K}]^\top, \end{aligned} \quad (6)$$

with vec being the vectorization operator, and $\tilde{\mathbf{X}} \in \mathbb{R}^{KC \times T}$ denotes the augmented signal matrix:

$$\tilde{\mathbf{X}} = [\mathbf{X}^\top, [\mathbf{X}^{(\tau)}]^\top, \dots, [\mathbf{X}^{((K-1)\tau)}]^\top]^\top, \quad (7)$$

where $\mathbf{X}^{(\eta)} \in \mathbb{R}^{C \times T}$ is $\mathbf{X}$ delayed by η sampled time points. By definition, $\mathbf{X}^{(0)} = \mathbf{X}$.

Therefore, optimizing the spatial filter $\mathbf{s}$ and the channel-wise temporal filter $\mathbf{a}_c$'s ($c \in \{1, 2, \dots, C\}$) is equivalent to optimizing the combined spatio-temporal filter $\mathbf{v}$. Once $\mathbf{v}$ is optimized, $\mathbf{s}$ and $\mathbf{a}_c$'s can be recovered by decomposing $\mathbf{v}$, which nonetheless are not unique. Let $\mathbf{v}_c = [v_{c,1}, v_{c,2}, \dots, v_{c,K}]^\top$ be the channel-wise spatio-temporal filter. In line with [12], we constrain each temporal filter to have unit l_2 -norm, and the sign of the corresponding weight in the spatial filter to be identical to that of $v_{c,1}$. Consequently, the c -th coefficient of $\mathbf{s}$ satisfies

$$s_c = \text{sgn}[v_{c,1}] \cdot \sqrt{(v_{c,1})^2 + (v_{c,2})^2 + \dots + (v_{c,K})^2}, \quad (8)$$

where $\text{sgn}[x] = 1$ when $x \geq 0$, and $\text{sgn}[x] = -1$ otherwise. The temporal filters can then be computed as

$$\mathbf{a}_c = \frac{1}{s_c} \cdot [v_{c,1}, v_{c,2}, \dots, v_{c,K}]^\top, \quad c \in \{1, 2, \dots, C\}. \quad (9)$$

C. Spatio-Temporal-Filtering-Based EEG Decoding as Linear Matrix Regression

In order to learn the optimal mapping f in (2), one is typically provided with a training set comprising multiple EEG trials $\{\mathbf{X}_m, y_m\}_{m=1}^M$, where M denotes the number of training samples; each $\mathbf{X}_m \in \mathbb{R}^{C \times T}$ denotes the m -th single-trial EEG signal, and $y_m \in \mathbb{R}$ the corresponding target variable.

With a set of L spatio-temporally filtered signals $\mathbf{z}_l = \tilde{\mathbf{X}}_m^\top \mathbf{v}_l$ ($l = 1, 2, \dots, L$), the target variable y_m is typically related to their spectral power features $\mathbf{p} = [\mathbf{z}_1^\top \mathbf{z}_1, \mathbf{z}_2^\top \mathbf{z}_2, \dots, \mathbf{z}_L^\top \mathbf{z}_L]^\top \in \mathbb{R}^{L \times 1}$ via a linear regression model with the weight vector $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_L]^\top \in \mathbb{R}^{L \times 1}$:

$$\begin{aligned} y_m &= f(\mathbf{X}_m) + \epsilon = \boldsymbol{\alpha}^\top \mathbf{p} + \epsilon \\ &= \sum_{l=1}^L \alpha_l \mathbf{z}_l^\top \mathbf{z}_l + \epsilon = \sum_{l=1}^L \alpha_l \mathbf{v}_l^\top \tilde{\mathbf{X}}_m \tilde{\mathbf{X}}_m^\top \mathbf{v}_l + \epsilon, \end{aligned} \quad (10)$$

where $\tilde{\mathbf{X}}_m$ is the augmented signal matrix of the m -th trial as defined in (7). It can be seen that even though the regression model is linear, $f(\mathbf{X}_m)$ is nonlinear with respect to the spatio-temporal filters $\mathbf{v}_l$ ($l = 1, 2, \dots, L$) due to the presence of their quadratic terms. Attempting to optimize the spatio-temporal filters and regression weights directly would likely result in local minima. This issue can be resolved by converting the above model to a linear one with respect to the unknown parameters via reparameterization. Specifically, by applying additional algebraic manipulations to (10) we have

$$y_m = \text{tr} \left[\sum_{l=1}^L \alpha_l \mathbf{v}_l \mathbf{v}_l^\top \tilde{\mathbf{X}}_m \tilde{\mathbf{X}}_m^\top \right] + \epsilon = \text{tr}[\mathbf{W} \tilde{\mathbf{R}}_m] + \epsilon, \quad (11)$$

where tr denotes the trace operator which computes the sum of the diagonal elements of a matrix, and $\tilde{\mathbf{R}}_m = \tilde{\mathbf{X}}_m \tilde{\mathbf{X}}_m^\top \in \mathbb{R}^{KC \times KC}$ denotes the empirical augmented covariance matrix (up to a constant factor of KC) of the EEG for the m -th trial. $\mathbf{W}$ is a symmetric matrix that reparameterizes the spatio-temporal filters and regression weights:

$$\mathbf{W} = \sum_{l=1}^L \alpha_l \mathbf{v}_l \mathbf{v}_l^\top. \quad (12)$$

Suppose $\mathbf{v}_l$'s are orthogonal unit vectors. Then (12) becomes the eigenvalue decomposition of $\mathbf{W}$, with α and $\mathbf{v}_l$'s being the eigenvalues and eigenvectors of $\mathbf{W}$. It should be noted that despite being nonlinear with respect to α and $\mathbf{v}_l$'s, f is a linear function of $\mathbf{W}$. As will be shown, this property is desirable as it is easier to optimize $\mathbf{W}$ than directly optimizing α and $\mathbf{v}_l$'s. Once $\mathbf{W}$ is optimized, α and $\mathbf{v}_l$'s can be obtained by performing the eigenvalue decomposition of $\mathbf{W}$ for interpretation purpose.

Specifically, $\mathbf{W}$ can be optimized by minimizing the following averaged squared linear loss:

$$\min_{\mathbf{W}} \frac{1}{M} \sum_{m=1}^M (y_m - \text{tr}[\mathbf{W} \tilde{\mathbf{R}}_m])^2. \quad (13)$$

However, $L = KC$ without any constraints over the parameter space. The resulting model can easily overfit the data as the dimensionality of the parameter space is high compared to the sample size of the training set. To address this issue, we penalize the number of latent signals by adding the rank of $\mathbf{W}$, i.e., L , as a regularization term in the loss function, yielding the following optimization problem:

$$\min_{\mathbf{W}} \frac{1}{M} \sum_{m=1}^M (y_m - \text{tr}[\mathbf{W} \tilde{\mathbf{R}}_m])^2 + \rho \cdot \text{rank}[\mathbf{W}], \quad (14)$$

where $\text{rank}[\mathbf{W}]$ denotes the rank of $\mathbf{W}$, and ρ is the regularization parameter trading off the averaged squared linear loss and $\text{rank}[\mathbf{W}]$.

Alternatively, denote the column-wise vectorization of each $\tilde{\mathbf{R}}_m$ as $\tilde{\mathbf{r}}_m = \text{vec}[\tilde{\mathbf{R}}_m] \in \mathbb{R}^{K^2 C^2 \times 1}$ ($m = 1, 2, \dots, M$), and a similar vectorization of $\mathbf{W}$ as $\tilde{\mathbf{w}} = \text{vec}[\mathbf{W}] \in \mathbb{R}^{K^2 C^2 \times 1}$. Moreover, let

$$\begin{aligned} \tilde{\mathbf{R}} &= [\tilde{\mathbf{r}}_1, \tilde{\mathbf{r}}_2, \dots, \tilde{\mathbf{r}}_M]^\top, \\ \mathbf{y} &= [y_1, y_2, \dots, y_M]^\top, \end{aligned} \quad (15)$$

then Problem (14) becomes

$$\min_{\mathbf{W}} \frac{1}{M} \|\mathbf{y} - \tilde{\mathbf{R}} \tilde{\mathbf{w}}\|_2^2 + \rho \cdot \text{rank}[\mathbf{W}]. \quad (16)$$

However, Problem (16) is NP-hard as $\text{rank}[\mathbf{W}]$ is a highly non-convex function of $\mathbf{W}$. Although replacing $\text{rank}[\mathbf{W}]$ with a convex surrogate $\|\mathbf{W}\|_* = \sum_{l=1}^L |\alpha_l|$ can yield a convex optimization problem, it may fail to recover the solution for Problem (16) under certain conditions [28]. Moreover, the regularization parameter ρ is often determined via cross-validation, which can be time-consuming and does not make efficient use of the data since a portion of the data are held out for testing purpose in each fold of the cross-validation.

D. Parameter and Hyperparameter Optimization Via Sparse Bayesian Learning

Having formulated EEG decoding as a rank-regularized linear matrix regression problem, we now develop a computationally efficient algorithm to solve the non-convex Problem (16) under a principled SBL framework, which enables us to reframe the optimization problem under a probabilistic setting. Importantly, this framework also allows us to determine the regularization

parameter via marginal likelihood maximization, which automatically embodies a notion of Occam's razor and thus obviates the need of out-of-sample model evaluation.

Below we first provide a probabilistic model where a parameterized Gaussian prior is placed on the weight matrix. It can be shown that maximizing the marginal likelihood associated with this model promotes a low rank of the weight matrix. We then derive update rules for iteratively optimizing the parameters and hyperparameters in the model.

1) *Probabilistic Modeling*: To recast Problem (16) in a Bayesian framework, we first define the following Gaussian likelihood function:

$$p(\mathbf{y}|\mathbf{W}; \vec{\mathbf{R}}, \lambda) \propto \exp \left[-\frac{1}{2\lambda} \|\mathbf{y} - \vec{\mathbf{R}}\vec{\mathbf{w}}\|_2^2 \right], \quad (17)$$

where λ denotes the unknown variance of the additive noise. Subsequently, a zero-mean Gaussian prior distribution with covariance matrix $\gamma_j \Psi$ is placed on $\mathbf{w}_j$ ($j = 1, 2, \dots, KC$), i.e., each column of the matrix $\mathbf{W}$:

$$p(\mathbf{w}_j; \Psi, \gamma) = \mathcal{N}(\mathbf{w}_j; \mathbf{0}, \gamma_j \Psi), \quad (18)$$

where $\mathcal{N}$ denotes the Gaussian distribution, $\Psi \in \mathbb{R}^{KC \times KC}$ is an unknown positive semi-definite and symmetric matrix, and $\gamma = [\gamma_1, \gamma_2, \dots, \gamma_{KC}]^T \in \mathbb{R}^{KC \times 1}$ is also unknown.

Since $\mathbf{W}$ is a symmetric matrix, each row of $\mathbf{W}$ follows the identical Gaussian prior as the corresponding column in $\mathbf{W}$. As such, let $\bar{\Psi}^{(c)} = \text{diag}(\gamma) \otimes \Psi$ and $\bar{\Psi}^{(r)} = \Psi \otimes \text{diag}(\gamma)$, with $\otimes$ denoting the Kronecker product. The prior on $\mathbf{W}$ can then be obtained as:

$$p(\mathbf{W}; \Psi, \gamma) \propto \exp \left[-\frac{1}{2} \vec{\mathbf{w}}^T \bar{\Psi}^{-1} \vec{\mathbf{w}} \right], \quad (19)$$

where $\bar{\Psi} = \frac{1}{2}(\bar{\Psi}^{(c)} + \bar{\Psi}^{(r)}) \in \mathbb{R}^{K^2 C^2 \times K^2 C^2}$.

Since both the likelihood and prior are Gaussian, the posterior $p(\mathbf{W}|\mathbf{y}; \vec{\mathbf{R}}, \lambda, \Psi, \gamma)$ is also Gaussian, with the vectorized mean $\vec{\mathbf{w}}$ given by¹

$$\vec{\mathbf{w}} = \text{vec}[\bar{\mathbf{W}}] = \bar{\Psi} \vec{\mathbf{R}}^T \Sigma_y^{-1} \mathbf{y}, \quad (20)$$

where $\mathbf{I}$ denotes the identity matrix,

$$\Sigma_y = \vec{\mathbf{R}} \bar{\Psi} \vec{\mathbf{R}}^T + \lambda \mathbf{I} = \sum_{j=1}^{KC} \gamma_j \vec{\mathbf{R}}_j \Psi \vec{\mathbf{R}}_j^T + \lambda \mathbf{I} \in \mathbb{R}^{M \times M}, \quad (21)$$

and $\vec{\mathbf{R}} = [\vec{\mathbf{R}}_1, \vec{\mathbf{R}}_2, \dots, \vec{\mathbf{R}}_{KC}]$, with $\vec{\mathbf{R}}_j \in \mathbb{R}^{M \times KC}$ ($j \in \{1, 2, \dots, KC\}$).

As a consequence,

$$\bar{\mathbf{W}} = \frac{1}{2}(\bar{\mathbf{W}}^{(c)} + [\bar{\mathbf{W}}^{(c)}]^T), \quad (22)$$

where $\bar{\mathbf{w}}^{(c)} = \text{vec}[\bar{\mathbf{W}}^{(c)}] = \mathbf{B} \Sigma_y^{-1} \mathbf{y}$, and $\mathbf{B} = \bar{\Psi}^{(c)} \vec{\mathbf{R}}^T = [\vec{\mathbf{R}}_1 \Psi^T, \vec{\mathbf{R}}_2 \Psi^T, \dots, \vec{\mathbf{R}}_{KC} \Psi^T]^T$. It can be seen that $\bar{\mathbf{W}}$ resides in a low-dimensional subspace if Ψ has a low rank, resulting in a low-rank $\bar{\mathbf{W}}$ as well.

¹ See Supplemental Material, available online for the derivation of $\bar{\mathbf{W}}$.

2) *Hyperparameter Optimization*: From an empirical Bayes perspective, Ψ , γ , and λ can be estimated by maximizing the marginal likelihood function $p(\mathbf{y}|\vec{\mathbf{R}}, \lambda, \Psi, \gamma)$ with $\mathbf{W}$ integrated out from the likelihood function:

$$\max_{\Psi \in \mathbb{H}^+, \lambda, \gamma_j \geq 0} \int p(\mathbf{y}|\mathbf{W}; \vec{\mathbf{R}}, \lambda) p(\mathbf{W}; \Psi, \gamma) d\mathbf{W}, \quad (23)$$

where $\mathbb{H}^+$ denotes the set of $KC \times KC$ symmetric positive semi-definite matrices.

Replacing $p(\mathbf{y}|\mathbf{W}; \vec{\mathbf{R}}, \lambda)$ and $p(\mathbf{W}; \Psi, \gamma)$ with (17) and Equation (19), respectively, with some algebraic manipulations Problem (23) can be reduced to the following problem:

$$\min_{\Psi \in \mathbb{H}^+, \lambda, \gamma_j \geq 0} \mathcal{L}(\Psi, \gamma, \lambda) \triangleq \mathbf{y}^T \Sigma_y^{-1} \mathbf{y} + \log[|\Sigma_y|]. \quad (24)$$

$\mathcal{L}(\Psi, \gamma, \lambda)$ comprises two terms: the first term $\mathbf{y}^T \Sigma_y^{-1} \mathbf{y}$ measures the discrepancy between the empirical covariance $\mathbf{y} \mathbf{y}^T$ and model covariance Σ_y ; the second term $\log[|\Sigma_y|]$ is the log-volume of the high-dimensional parallelepiped formed by Σ_y . Intuitively, the volume of a high-dimensional parallelepiped can be reduced by making the parallelepiped nearly collapse over many dimensions, which amounts to lowering the rank of Ψ based on (21). Therefore, minimizing $\mathcal{L}(\Psi, \gamma, \lambda)$ offers a trade-off between fit of the model to the class labels and the rank of the Ψ , the latter inducing sparsity in the EEG latent space.

3) *Optimization Algorithm*: For simplicity, we assume $\gamma = \mathbf{1} = [1, \dots, 1]^T \in \mathbb{R}^{KC \times 1}$, reducing $\mathcal{L}(\Psi, \gamma, \lambda)$ to $\mathcal{L}(\Psi, \lambda) \triangleq \mathcal{L}(\Psi, \gamma = \mathbf{1}, \lambda)$. However, we note this assumption is not essential for the algorithm to work and the more general setting is still tractable and will be studied in the future.

One can employ a general-purpose nonlinear optimization algorithm to solve Problem (24) as a non-convex constrained optimization problem. Here we instead develop an iterative algorithm that leverages dual-space representations to upper bound $\mathcal{L}(\Psi, \lambda)$ for facilitating optimization. This results in an EM-like algorithm with updates naturally satisfying the positive semi-definite constraint on Ψ and λ .

Lemma 1: An alternative representation of the first term of $\mathcal{L}(\Psi, \lambda)$ is as follows:

$$\mathbf{y}^T \Sigma_y^{-1} \mathbf{y} = \min_{\mathbf{u}} \lambda^{-1} \|\mathbf{y} - \vec{\mathbf{R}} \mathbf{u}\|_2^2 + \mathbf{u}^T \bar{\Psi}^{-1} \mathbf{u}, \quad (25)$$

where $\mathbf{u} \in \mathbb{R}^{K^2 C^2 \times 1}$.

Let

$$\mathbf{u}^{opt} = \arg \min_{\mathbf{u}} \lambda^{-1} \|\mathbf{y} - \vec{\mathbf{R}} \mathbf{u}\|_2^2 + \mathbf{u}^T \bar{\Psi}^{-1} \mathbf{u},$$

it can be shown that

$$\mathbf{u}^{opt} = \vec{\mathbf{w}} = \bar{\Psi} \vec{\mathbf{R}}^T \Sigma_y^{-1} \mathbf{y}. \quad (26)$$

Moreover, using the matrix determinant lemma, the second term of $\mathcal{L}(\Psi, \lambda)$ can be represented as

$$\begin{aligned} \log[|\Sigma_y|] &= \log[|\vec{\mathbf{R}} [\bar{\Psi}^{(c)}]^T \vec{\mathbf{R}}^T + \lambda \mathbf{I}|] \\ &= \log[\lambda^{-1} \vec{\mathbf{R}}^T \vec{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}] \\ &\quad + \log[|\bar{\Psi}^{(c)}|] + M \log[\lambda]. \end{aligned} \quad (27)$$

Let

$$\Delta = \begin{bmatrix} \Psi^{-1} & \mathbf{0} \\ \mathbf{0} & \lambda^{-1} \end{bmatrix}, \quad (28)$$

then we have

Lemma 2: $\log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}]$ is concave with respect to Δ .

The fact that $\log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}]$ is concave with respect to Δ admits the following dual representation:

$$\log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}] = \min_{\Omega} \text{tr}[\Delta\Omega] + h^*(\Omega), \quad (29)$$

where

$$\Omega = \begin{bmatrix} \Phi & \mathbf{0} \\ \mathbf{0} & \theta \end{bmatrix} \quad (30)$$

is the dual variable, and $h^*(\Omega)$ is the conjugate function of $\log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}]$ independent of Δ .

Theorem 3: Problem (24) is equivalent to

$$\min_{\Psi, \lambda, \Omega, \mathbf{u}} \mathcal{L}(\Psi, \mathbf{u}, \Omega, \lambda), \quad (31)$$

where

$$\begin{aligned} \mathcal{L}(\Psi, \mathbf{u}, \Omega, \lambda) = & \lambda^{-1}\|\mathbf{y} - \tilde{\mathbf{R}}\mathbf{u}\|_2^2 + \mathbf{u}^\top \bar{\Psi}^{-1}\mathbf{u} \\ & + M \log[\lambda] + \text{tr}[\Psi^{-1}\Phi] + \lambda^{-1}\theta \\ & + h^*(\Omega) + KC \log[\|\Psi\|]. \end{aligned} \quad (32)$$

The proofs of Lemmas 1, 2 and Theorem 3 are provided in Supplemental Material, available online. Ψ , λ can therefore be optimized along with Ω and $\mathbf{u}$ in an alternate fashion via the coordinate descent.

Let

$$\Omega^{opt} = \arg \min_{\Omega} \text{tr}[\Delta\Omega] + h^*(\Omega). \quad (33)$$

Geometrically determining Ω^{opt} amounts to finding the hyperplane tangential to $\log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}]$ at each Δ . Therefore,

$$\Omega^{opt} = \nabla_{\Delta} \log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}]. \quad (34)$$

Given that Δ and Ω are diagonal block matrices, obtaining Ω^{opt} is equivalent to calculating the gradients with respect to Ψ^{-1} and λ^{-1} , resulting in Φ^{opt} and θ^{opt} respectively. These optimal values can then be combined to yield:

$$\Omega^{opt} = \begin{bmatrix} \Phi & \mathbf{0} \\ \mathbf{0} & \theta \end{bmatrix}, \quad (35)$$

with

$$\begin{aligned} \Phi^{opt} &= \nabla_{\Psi^{-1}} \log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}] \\ &= \sum_{j=1}^{KC} (\Psi - \Psi\tilde{\mathbf{R}}_j^\top \Sigma_y^{-1} \tilde{\mathbf{R}}_j \Psi), \end{aligned} \quad (36)$$

$$\theta^{opt} = \nabla_{\lambda^{-1}} \log[\lambda^{-1}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} + [\bar{\Psi}^{(c)}]^{-1}]$$

$$= \sum_{j=1}^{KC} \text{tr}[\Psi\tilde{\mathbf{R}}_j^\top \tilde{\mathbf{R}}_j - \Psi\tilde{\mathbf{R}}_j^\top \Sigma_y^{-1} \tilde{\mathbf{R}}_j \Psi\tilde{\mathbf{R}}_j^\top \tilde{\mathbf{R}}_j]. \quad (37)$$

Given γ , $\mathbf{u}$, θ and Φ , the update rule of Ψ can be obtained by setting $\frac{\partial \mathcal{L}(\Psi, \mathbf{u}, \Phi, \theta)}{\partial \Psi} = 0$:

$$\Psi^{opt} = \frac{\mathbf{U}\mathbf{U}^\top + \Phi}{KC}, \quad (38)$$

where $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{KC}]$ with each column satisfies $[\mathbf{u}_1^\top, \mathbf{u}_2^\top, \dots, \mathbf{u}_{KC}^\top]^\top = \mathbf{u}$, $\mathbf{u}_j \in \mathbb{R}^{KC \times 1}$ ($j \in \{1, 2, \dots, KC\}$).

Likewise, the update rule of λ can be derived by setting $\frac{\partial \mathcal{L}(\Psi, \mathbf{u}, \Phi, \theta)}{\partial \lambda} = 0$:

$$\lambda^{opt} = \frac{\|\mathbf{y} - \tilde{\mathbf{R}}\mathbf{u}\|_2^2 + \theta}{M}. \quad (39)$$

With Ψ and λ initialized to $\mathbf{I}$ and 1, respectively, $\mathbf{u}$, Φ , Ψ , θ , and λ can be iteratively updated via Eqs. (26), (36), (38), (37), and (39) until convergence, which is guaranteed by reaching a stationary point of the cost function [28]. We can then set $\mathbf{W} = \mathbf{U}$ upon convergence.

The spatio-temporal filters and the classifier weights can be obtained via the eigenvalue decomposition of $\mathbf{W}$ as in Equation (12). In order to numerically determine the rank of $\mathbf{W}$, namely L , the eigenvalues of $\mathbf{W}$ are normalized by the maximum eigenvalue:

$$\bar{\alpha}_j = \frac{\alpha_j}{\max\{|\alpha_1|, |\alpha_2|, \dots, |\alpha_{KC}|\}}. \quad (40)$$

Then L is set to the number of $\bar{\alpha}_j$'s that exceed a pre-defined threshold, e.g., 0.05.

The pseudocode for this algorithm, termed SBLEST, is provided in Algorithm 1.

III. EXPERIMENTS AND RESULTS

To validate the efficacy of the proposed SBLEST algorithm, we systematically assess its decoding performance on multiple MI BCI and emotion recognition datasets. In addition, our algorithm is compared against several contemporary algorithms, including CSP-based algorithms, nuclear-norm-based sparse learning algorithms, and state-of-the-art deep learning algorithms. We also illustrate the interpretability of our model by visualizing the extracted spatio-temporal patterns in relation to the underlying neurophysiology.

A. EEG Datasets

Five MI BCI datasets and one emotion recognition dataset are employed to assess the binary classification performance of different algorithms. For the MI BCI datasets, three are from past BCI competitions, one is a dataset from Tsinghua University, and one is the OpenBMI dataset [32]. For emotion recognition, the SJTU Emotion EEG Dataset (SEED) is analyzed. Details for all the datasets are provided below, with the electrode montages depicted in Supplemental Material, available online.

1) *Dataset I* (Dataset IVa from BCI Competition III [33]; $N = 5$): This dataset contains EEG signals recorded from 5 healthy subjects: *al*, *aa*, *av*, *aw*, and *ay*. The subjects performed a classic

Algorithm 1: SBLEST.

Input: EEG data of C channels, T sampled time points, and M training samples $\mathbf{X}_m \in \mathbb{R}^{C \times T}$, $y_m \in \{-1, 1\}$ ($m \in \{1, 2, \dots, M\}$); temporal filter order K ; delay time τ , test sample $\mathbf{X}_{te} \in \mathbb{R}^{C \times T}$.

Output: Low-rank matrix $\mathbf{W}$; spatio-temporal filters $\mathbf{v}_l$ and classifier weights α_l ($l \in \{1, 2, \dots, L\}$), predicted label $\hat{y}_{te}$.

1: **Initialization:** $\Psi = \mathbf{I}$, $iter = 1$, $\lambda = 1$, $\mathcal{L}(1) = 10^{12}$.

2: $\tilde{\mathbf{X}}_m = [\mathbf{X}_m^\top, \mathbf{X}_m^{(\tau)\top}, \dots, \mathbf{X}_m^{((K-1)\tau)\top}]^\top$

3: $\tilde{\mathbf{R}}_m = \tilde{\mathbf{X}}_m \tilde{\mathbf{X}}_m^\top$

4: $\tilde{\mathbf{r}}_m = \text{vec}[\tilde{\mathbf{R}}_m] \in \mathbb{R}^{K^2 C^2 \times 1}$

5: $\tilde{\mathbf{R}} = [\tilde{\mathbf{r}}_1, \tilde{\mathbf{r}}_2, \dots, \tilde{\mathbf{r}}_M]^\top$

6: $\mathbf{y} = [y_1, y_2, \dots, y_M]^\top$

7: $[\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2, \dots, \tilde{\mathbf{R}}_{KC}] = \tilde{\mathbf{R}}$, where $\tilde{\mathbf{R}}_j \in \mathbb{R}^{M \times KC}$ ($j \in \{1, 2, \dots, KC\}$)

8: **while** 1 **do**

9: $\mathbf{B} = [\tilde{\mathbf{R}}_1 \Psi^\top, \tilde{\mathbf{R}}_2 \Psi^\top, \dots, \tilde{\mathbf{R}}_{KC} \Psi^\top]^\top$

10: $\Sigma_y = \sum_{j=1}^{KC} \tilde{\mathbf{R}}_j \Psi \tilde{\mathbf{R}}_j^\top + \lambda \mathbf{I}$

11: $\mathbf{u}^{(c)} = \mathbf{B} \Sigma_y^{-1} \mathbf{y}$

12: $[[\mathbf{u}_1^{(c)}]^\top, [\mathbf{u}_2^{(c)}]^\top, \dots, [\mathbf{u}_{KC}^{(c)}]^\top]^\top = \mathbf{u}^{(c)}$, where $\mathbf{u}_j^{(c)} \in \mathbb{R}^{KC \times 1}$, ($j \in \{1, 2, \dots, KC\}$)

13: $\mathbf{U}^{(c)} = [\mathbf{u}_1^{(c)}, \mathbf{u}_2^{(c)}, \dots, \mathbf{u}_{KC}^{(c)}]$

14: $\mathbf{U} = \frac{\mathbf{U}^{(c)} + [\mathbf{U}^{(c)}]^\top}{2}$

15: $\mathbf{u} = \text{vec}[\mathbf{U}]$

16: $\Phi_j = \Psi - \Psi \tilde{\mathbf{R}}_j^\top \Sigma_y^{-1} \tilde{\mathbf{R}}_j \Psi$

17: $\Phi = \sum_{j=1}^{KC} \Phi_j$

18: $\Psi = \frac{\mathbf{U} \mathbf{U}^\top + \Phi}{KC}$

19: $\theta = \sum_{j=1}^{KC} \text{tr}[\Phi_j \tilde{\mathbf{R}}_j^\top \tilde{\mathbf{R}}_j]$

20: $\lambda = \frac{\|\mathbf{y} - \tilde{\mathbf{R}} \mathbf{u}\|_2^2 + \theta}{M}$

21: $iter = iter + 1$

22: $\mathcal{L}(iter) = \mathbf{y}^\top \Sigma_y^{-1} \mathbf{y} + \log[\|\Sigma_y\|]$

23: $\Delta \mathcal{L} = \frac{|\mathcal{L}(iter-1) - \mathcal{L}(iter)|}{|\mathcal{L}(iter-1)|}$

24: **if** $iter > 5000$ or $\Delta \mathcal{L} < 2 \times 10^{-4}$ **then**

25: **break**

26: **end if**

27: **end while**

28: $\mathbf{W} = \mathbf{U}$

29: Eigenvalue Decomposition of $\mathbf{W}$:

$$\mathbf{W} = \sum_{l=1}^L \alpha_l \mathbf{v}_l \mathbf{v}_l^\top$$

30: $\tilde{\mathbf{X}}_{te} = [\mathbf{X}_{te}^\top, \mathbf{X}_{te}^{(\tau)\top}, \dots, \mathbf{X}_{te}^{((K-1)\tau)\top}]^\top$

31: $\tilde{\mathbf{R}}_{te} = \tilde{\mathbf{X}}_{te} \tilde{\mathbf{X}}_{te}^\top$

32: $\hat{y}_{te} = \text{sgn}[\text{tr}[\mathbf{W}^\top \tilde{\mathbf{R}}_{te}]]$

cue-based motor imagery paradigm with two tasks (right-hand and foot, labeled 1 and 2). EEG signals were recorded using 118 channels with a sampling rate of 100 Hz. The experiment was repeated for 280 trials for each subject, among which 224, 168, 84, 56 and 28 trials composed the training set for subject *al*, *aa*, *av*, *aw*, and *ay* respectively, and the remaining trials composed the testing set.

2) *Dataset II* (Dataset IIIa from BCI Competition III [34]; $N = 18$): This dataset contains EEG signals recorded from 3 healthy subjects, i.e., *K3*, *K6*, and *L1*. The subjects performed four motor imagery tasks (left-hand, right-hand, foot, and tongue, labeled 1, 2, 3, and 4). EEG signals were recorded using 60 channels with a sampling rate of 250 Hz. Both the training and test sets contain 45 trials per class for *K3*, and

30 trials per class for *K6* and *L1*. For the purpose of binary classification, $C_4^2 = 6$ class pairs are generated for each subject, resulting in 18 data subsets. Henceforth we refer to each data subset as “subject ID-labels” for convenience.

3) *Dataset III* (Dataset IIa from BCI Competition IV [35]; $N = 54$): The dataset is comprised of 22-channel EEG signals from 9 subjects (subjects A01-A09) performing left-hand, right-hand, foot, and tongue motor imageries, with a sampling rate of 250 Hz. Both the training and testing sets contain 72 trials per class for each subject. As with Dataset II, the four-class data of each subject is split into $C_4^2 = 6$ data subsets, yielding $6 \times 9 = 54$ data subsets.

4) *Dataset IV* (The Tsinghua dataset [36]; $N = 7$): The in-house dataset is provided by Tsinghua University, which contains EEG signals recorded from 7 healthy subjects (*S1-S7*). The subjects performed left-hand and right-hand motor imagery experiment tasks (labeled 1 and 2) for several sessions without feedback. The EEG signals were recorded using 32 channels with a sampling rate of 256 Hz. In total, 300 trials for Sub1-Sub4 and 240 trials for Sub5-Sub7 are available with balanced labels. The training set is composed of 60 trials (30 trials per class) recorded in a single session, and the remaining 240 trials for Sub1-Sub4 and 180 trials for Sub5-Sub7 compose the test datasets.

5) *Dataset V* (The OpenBMI dataset [32]; $N = 108$): This dataset is composed of EEG signals recorded from 54 subjects, each subject performed two sessions. The experiment in each session consisted of training and test phases, and each phase had 100 trials with balanced right-hand and left-hand imagery tasks. The EEG signals were recorded using 62 channels with a sampling rate of 250 Hz. In this work, each session is treated as a subject, thus 108 data subsets (*SUB1-SUB108*) are obtained for binary classification evaluation.

6) *Dataset VI* (The SEED dataset [16]; $N = 45$): The SEED dataset is comprised of 62-channel EEG signals from 15 subjects (*SUB1-SUB15*), which were recorded when they were watching Chinese film clips with three types of emotions i.e., negative, positive, and neutral. The data collection was performed in 3 sessions, each containing 15 film clips with 5 approximately 4-minute-duration clips for each emotion. The EEG data was then downsampled to 200 Hz. In our binary classification experiment, the EEG in each session was treated as a data subset, resulting in a total of 45 EEG data subsets. Moreover, only the positive and negative emotions were considered, such that 10 out of 15 film clips were used for each data subset. The EEG data of each film clip was cut into non-overlapping 5-s samples. As a result, the number of samples in each data subset was 455. Subsequently, we divided each data subset into 5 equal-sized folds in chronological order, each consisting of a positive and a negative film clips. Among them, the first three folds were used as the training set, and the remaining two folds were used as the test set. Note this particular split ensures that there is no leakage of the data from the same film clips into both the training and test sets, which would inflate the prediction performance.

In summary, $5 + 18 + 54 + 7 + 108 = 192$ MI BCI data subsets and 45 emotion recognition data subsets were used for within-subject performance evaluation.

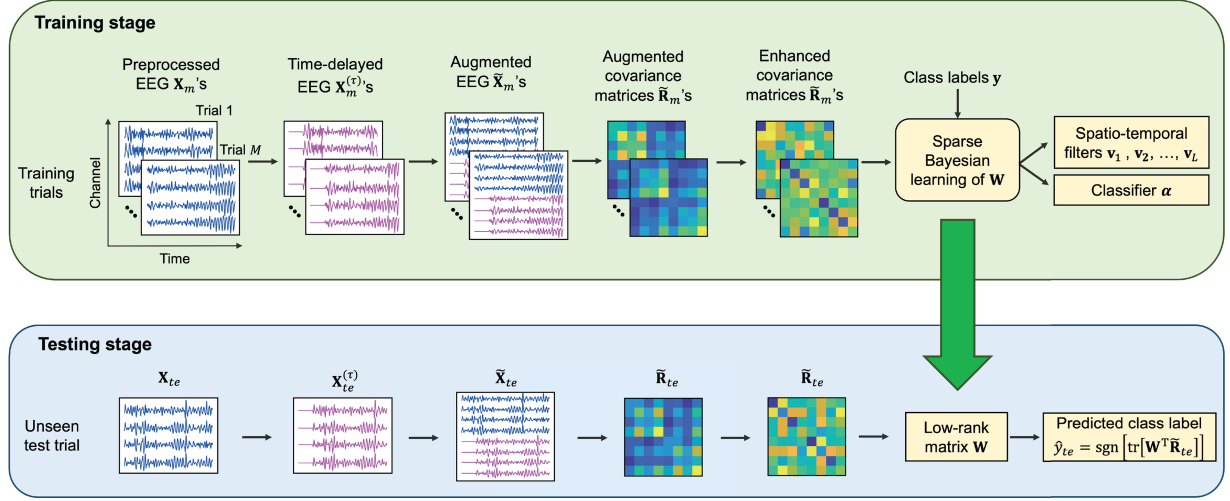


Fig. 1. Analysis procedure for SBLEST. $\mathbf{X}_m$ ($m = 1, 2, \dots, M$) and $\mathbf{X}_t$ denote the training and test EEG data, respectively. The enhancement steps applied to the augmented covariance matrices consist of trace normalization, whitening, and matrix logarithm.

TABLE I
HYPERPARAMETERS AND SEARCH SPACE FOR CSSP, CSSSP, LRDS, SVM, DBN, GELM, AND SBLEST

Algorithm	Hyperparameter	Search space
CSSP	delay time τ	$\tau \in \{0, 1, 2, 3, 4, 5\}$
CSSSP	regularization parameter t	$t \in \{0.1, 0.2, \dots, 5\}$
	order of FIR filter K	$K = 6$
LRDS	regularization parameter ρ	$\rho = 10^\alpha, \alpha \in \{-3, -2.8, \dots, -1.2, -1\}$
SVM	regularization parameter c	$c = 2^\alpha, \alpha \in \{-10, -9, \dots, 9, 10\}$
DBN	number of neurons in the first and second layers n_1, n_2	$n_1 \in \{200, 250, \dots, 500\}, n_2 \in \{100, 150, \dots, 500\}$
GELM	regularization parameters b_1, b_2	$b_1, b_2 = 10^\alpha, \alpha \in \{-3, -2, \dots, 3\}$
SBLEST	delay time τ	$\tau \in \{0, 1, 2, 3, 4, 5\}$
	order of FIR filter K	when $\tau = 0, K = 1$, otherwise $K = 2$

B. Data Analysis Procedures

As depicted in Fig. 1, model optimization was performed on the training set of each data subset. The hyperparameters were determined via grid search with 10-fold cross-validation for the MI BCI datasets and 3-fold cross-validation stratified over film clips for the SEED dataset (See Table I for the hyperparameters involved in different algorithms and the search space for each hyperparameter). The optimized models were applied to the test set to predict labels for the test samples. The classification accuracies, F1 scores [37] and Information Transfer Rates (ITRs) [38] were then computed to assess the decoding performance of different algorithms.

Specifically, SBLEST was compared to the following algorithms on the MI BCI datasets:

- *CSP-based algorithms*: CSP [8], FBCSP [13], CSSP [10], or CSSSP [11] for spatial or spatio-temporal filtering, followed by the FLDA for classification. Here FLDA was employed as the classifier due to its simplicity and excellent single-trial classification performance reported by previous BCI studies;
- *End-to-end algorithms*: logistic regression with dual spectrum norm regularizer (LRDS) [17], sCNN [21],

dCNN [21], EEGNet [22], EEG-inception [23], and EEGSym [24].

For the SEED dataset (Dataset VI), SBLEST was compared to classification methods utilizing the DE as the feature, defined as follows for a Gaussian distributed signal $\mathbf{S} \sim \mathcal{N}(\mu, \sigma^2)$:

$$h(\mathbf{S}) = - \int_{-\infty}^{+\infty} g(\mathbf{S}) \log[g(\mathbf{S})] dx = \frac{1}{2} \log[2e\pi\sigma^2], \quad (41)$$

where $g(\mathbf{S}) = \mathcal{N}(\mu, \sigma^2)$. The superiority of the DE feature was demonstrated in the SEED dataset [3], [16]. In line with these prior studies, three representative classifiers, including SVM, GELM, and DBN, were considered for classifying the DE features. It is worth noting that DE as defined in (41) is proportional to the logarithmic power of a signal, thus when applied to a specific frequency band the DE feature is equivalent to the logarithmic spectral band power feature.

To further constrain the model for SBLEST, we set $K = 2$ and allowed τ to vary from 0 to 5, which included the case when no temporal filtering was performed ($\tau = 0$). Moreover, the following steps empirically found to be beneficial to the classification performance were performed to enhance the empirical covariance matrix including the empirical augmented covariance matrix in CSP, FBCSP, CSSP, CSSSP, LRDS, and SBLEST:

(1) *Trace normalization*: In order to deal with the variability of the global EEG amplitude across trials, the empirical covariance matrices were normalized on an individual trial basis by dividing their own traces [4]. In particular, the trace normalization for SBLEST is as follows:

$$\tilde{\mathbf{R}}_m / \text{tr}[\tilde{\mathbf{R}}_m] \rightarrow \tilde{\mathbf{R}}_m (m = 1, 2, \dots, M). \quad (42)$$

(2) *Whitening*: Weak brain activity may be weighted unfavorably in the spatial/spatio-temporal filters when the discrimination between classes is quantified with distance measures. To address this limitation, a whitening step was applied to the empirical covariance matrices such that all activities were placed on the same footing with the identical variance [8]. In particular, for SBLEST the empirical augmented covariance matrix $\tilde{\mathbf{R}}_m$ was whitened as follows:

$$\mathbf{R}_p^{-\frac{1}{2}} \tilde{\mathbf{R}}_m \mathbf{R}_p^{-\frac{1}{2}} \rightarrow \tilde{\mathbf{R}}_m (m = 1, 2, \dots, M), \quad (43)$$

where $\mathbf{R}_p$ is the average of the empirical augmented covariance matrix across trials in the training set such that there is no data leakage from the test set into the training stage, and $\mathbf{R}_p^{-\frac{1}{2}}$ is the whitening matrix that can be derived from the Cholesky decomposition.

(3) *Logarithm transform*: Motivated by the fact that the logarithm of the spectral power feature is more Gaussian than the spectral power feature, the principal matrix logarithm was also applied to the empirical covariance matrices [39]. In particular, for SBLEST the empirical augmented covariance matrix was transformed as follows:

$$\text{logm}[\tilde{\mathbf{R}}_m] \rightarrow \tilde{\mathbf{R}}_m (m = 1, 2, \dots, M), \quad (44)$$

where $\text{logm}[\cdot]$ is the principal logarithm of a matrix.

1) *Data Analysis Procedure for CSP-Based Algorithms in MI Experiments*: CSP, FBCSP, CSSP and CSSSP were run on the 192 MI BCI data subsets for single-trial EEG classification, with the details provided below:

(1) *Preprocessing*:

- Channel selection: all channels were used for Dataset II, III and IV. For Dataset I and V, 49 channels and 20 channels were used for analysis as suggested in [40] and [32] respectively (EEG montages and selected channels are provided in Figs. S1 and S2 in Supplemental Material, available online).
- Bandpass filtering: 7-30 Hz band-pass filtering was performed using a 6-th order Butterworth filter. For FBCSP, the EEG signals were filtered into six sub-band components by six 6th order Butterworth filters (7-11 Hz, 11-15 Hz, 15-19 Hz, 19-23 Hz, 23-27 Hz, and 27-30 Hz).
- Time window selection: 0.5-3.5 s time segments were extracted, with 0 s indicating the time when the cue ended.

(2) *Feature extraction*: For each data subset, the preprocessed training data were fed into CSP, FBCSP, CSSP and CSSSP to optimize multiple spatial/spatio-temporal filters. The covariance matrices for CSP and FBCSP or augmented covariance matrices for CSSP and CSSSP were enhanced via trace normalization, whitening, and logarithm transform. The filters corresponding to the 3 largest eigenvalues for each class were chosen as

recommended in [8]. In order to enhance the Gaussianity of the features, the log-variance of the 6 spatially/spatio-temporally filtered signals were determined as the features. For FBCSP, 36 features were obtained as 6 features were optimized per sub-band.

(3) *Classification*: The features of the training set were used to train the FLDA. During the testing stage, features were fed into the FLDA to obtain the predicted labels.

2) *Data Analysis Procedure for End-to-End Algorithms in MI Experiments*: For the end-to-end algorithms, the preprocessing steps (i.e., channel selection, bandpass filtering, and time window selection) are identical to those for the CSP-based algorithms, with the subsequent downstream analyses described below:

- 1) For SBLEST, a flow chart is shown in Fig. 1. Specifically, the augmented data matrices $\tilde{\mathbf{X}}_m$ were first constructed from the preprocessed EEG data as in Equation (7). The empirical augmented covariance matrices were then calculated as $\tilde{\mathbf{R}}_m = \tilde{\mathbf{X}}_m \tilde{\mathbf{X}}_m^\top$ followed by trace normalization, whitening, and logarithm transform. Subsequently, with $\tilde{\mathbf{R}}_m$'s and the corresponding labels y_m 's in the training set, the low-rank $\mathbf{W}$ was optimized via SBL. $\mathbf{W}$ was then applied to each test EEG sample $\mathbf{X}_{te}$ to predict its class label as $\hat{y}_{te} = \text{sgn}[\text{tr}[\mathbf{W}^\top \tilde{\mathbf{R}}_{te}]]$. Finally, a set of spatial filters and channel-specific temporal filters, and the classifier can be obtained via the eigenvalue decomposition of $\mathbf{W}$ for interpretation and visualization purpose.
- 2) For deep learning algorithms, i.e., sCNN, dCNN, EEGNet, EEG-inception, and EEGSym, the preprocessed EEG and corresponding labels were used as inputs. To avoid overfitting, early stopping, batch normalization, and dropout were performed for all compared algorithms. The length of convolution kernels, as well as the size and stride of the pooling layers along the temporal dimension were proportionally adjusted based on the relationship between the EEG sampling rate and these hyperparameters determined in [21], [22]. Specifically, The convolutional kernel length of the first layer in sCNN, dCNN, and EEGNet corresponds to time windows of 100 ms, 40 ms, and 500 ms, respectively. Similarly, the convolutional kernel length in the inception Module of EEGSym and EEG-inception correspond to different temporal windows of duration 500 ms, 250 ms, and 125 ms. The network parameters used for sCNN, dCNN, and EEGNet in our study are presented in Table II, while the network parameters for EEG-inception and EEGSym can be found in [23], [24]. We used the open-source implementation of all deep learning algorithms provided in their respective original papers. The detailed training parameter settings for these models can be found in Table S1.
- sCNN, dCNN: sCNN is a shallow neural network with a temporal convolution layer, a frequency-specific spatial convolution layer, a squared max pooling layer, and a full connection layer. dCNN extends sCNN by adding three standard convolution-max-pooling blocks before the full connection layer. It was originally shown that cropped training improved the accuracy of dCNN significantly, but

TABLE II
NETWORK STRUCTURES AND HYPERPARAMETERS FOR SCNN, DCNN, AND EEGNET

Network	Block 1			Block 2			Classifier Layer
	Layer	kernels/strides	kernel size	Layer	kernels/strides	kernel size	
sCNN	Input ($C \times T$) temporal Conv2D spatial Conv2D square max pooling	40 40 $6f_s/100$	$(1 \times f_s/10)$ $(40 \times C)$ $(1 \times 3f_s/10)$				Dense Softmax
dCNN	Input ($C \times T$) temporal Conv2D spatial Conv2D max pooling	25 25 p	$(1 \times f_s/25)$ $(25 \times C)$ $(1 \times p)$	Conv2D max pooling Conv2D max pooling Conv2D max pooling	50 p 100 p 200 p	$(25 \times f_s/25)$ $(1 \times p)$ $(50 \times f_s/25)$ $(1 \times p)$ $(100 \times f_s/25)$ $(1 \times p)$	Dense Softmax
EEGNet	Input ($C \times T$) Conv2D DepthwiseConv2D Average pooling	8 2×8 $f_s/32$	$(1 \times f_s/2)$ $(C \times 1)$ $(1 \times f_s/32)$	SeparableConv2D Average pooling	16 $f_s/16$	$(1 \times f_s/8)$ $(1 \times f_s/16)$	Dense Softmax

C , T , and f_s denote the number of channels, the number of time points, and the sampling rate, respectively. p denotes the strides of the max pooling layer of dCNN. $p = 2$ for dataset 1, and $p = 3$ for the remaining datasets. Note that sCNN has only one block of spatio-temporal filters.

not for sCNN [21], hence cropped training was exclusively applied to the dCNN. Our study used the code provided at <https://github.com/robintibor/braindecode>.

- EEGNet: EEGNet is a deep learning architecture that leverages three 2-D convolution layers and a dense softmax classification layer to extract frequency-specific spatial features from EEG data and effectively integrate them. The model used in this study employs eight temporal filters and two spatial filters as shown in Table II. Our study used the code provided at <https://github.com/vlawhern/arl-eegmodels>.
- EEG-inception: EEG-Inception is the first model that incorporates inception modules for event-related potential (ERP) detection, achieving superior performance while maintaining a lightweight architecture. We adopted the same hyperparameters used in the original study for our motor imagery (MI) classification task [23]. To train and test our models, we used the open-source implementation of EEG-Inception, which is available at <https://github.com/esantamariavazquez/EEG-Inception>.
- EEGSym: This CNN leverages inception modules, residual connections, and a novel design that incorporates the mid-sagittal plane symmetry of the brain into the network architecture. The model was implemented using the code provided at <https://github.com/Serpeve/EEGSym>, and we followed the parameter settings recommended in [24], which also included employing data augmentation during the experiments as suggested.

To investigate the impact of various preprocessing steps on the performance of deep learning methods, we performed additional experiments for more comparisons. Specifically, the same preprocessing steps, including trace normalization and whitening, as in SBLEST were employed for deep learning methods.

Unlike SBLEST and the CSP-based methods that relied on the covariance matrix as input, deep learning methods utilize the EEG data matrix directly. As such, trace normalization and whitening could be equivalently applied to the EEG data matrix in the following manner:

$$\mathbf{X}_m / \sqrt{\text{tr}[\mathbf{R}_m]} \rightarrow \mathbf{X}_m (m = 1, 2, \dots, M), \quad (45)$$

$$\mathbf{R}_p^{-\frac{1}{2}} \mathbf{X}_m \rightarrow \mathbf{X}_m (m = 1, 2, \dots, M), \quad (46)$$

where $\mathbf{R}_m = \mathbf{X}_m \mathbf{X}_m^\top$ is the covariance matrix and $\mathbf{R}_p$ represents the average of $\mathbf{R}_m$ across trials in the training set.

3) Data Analysis Procedure for Emotion Recognition Dataset (Dataset VI)

1) Preprocessing

- Channel selection: all channels were utilized in our analysis.
- Bandpass filtering: 1-50 Hz band-pass filtering were performed using a 6th order Butterworth filter.

2) *Feature Extraction*: For SBLEST, the same analysis procedure as in Fig. 1 was employed. For the compared methods, the DE feature was computed per sample based on Equation (41) for each channel and each of the following five frequency bands: 1) delta: 1–4 Hz; 2) theta: 4–8 Hz; 3) alpha: 8–14 Hz; 4) beta: 14–31 Hz; and 5) gamma: 31–50 Hz. This resulted in a 310-dimension DE feature vector for each sample. A z-score normalization was then performed on the feature vector as suggested in [41].

3) *Classification*: The DE features of the training set were used to train three classifiers, namely SVM, DBN, and GELM. We use LIBSVM software to implement the SVM classifier and employ a linear kernel, a two-hidden-layer DBN constructed with the optimal numbers of neurons in the first and the second hidden layers selected via grid search. Their prediction performance was then compared with that of SBLEST on the test set.

4) *Cross-Subject Analysis Procedure for OpenBMI Dataset (Dataset V)*: Cross-subject analysis was conducted on Dataset V (number of subjects = 54) only, since the numbers of subjects in the other datasets (≤ 15) are insufficient for learning meaningful cross-subject models.

1) Preprocessing

- All data preprocessing steps were identical to those in the within-subject experiment, except that the data whitening step was discarded as it is unlikely that a common whitening matrix was shared among subjects.

2) Performance evaluation and hyperparameter selection

- Leave-one-subject-out strategy was employed for performance evaluation.

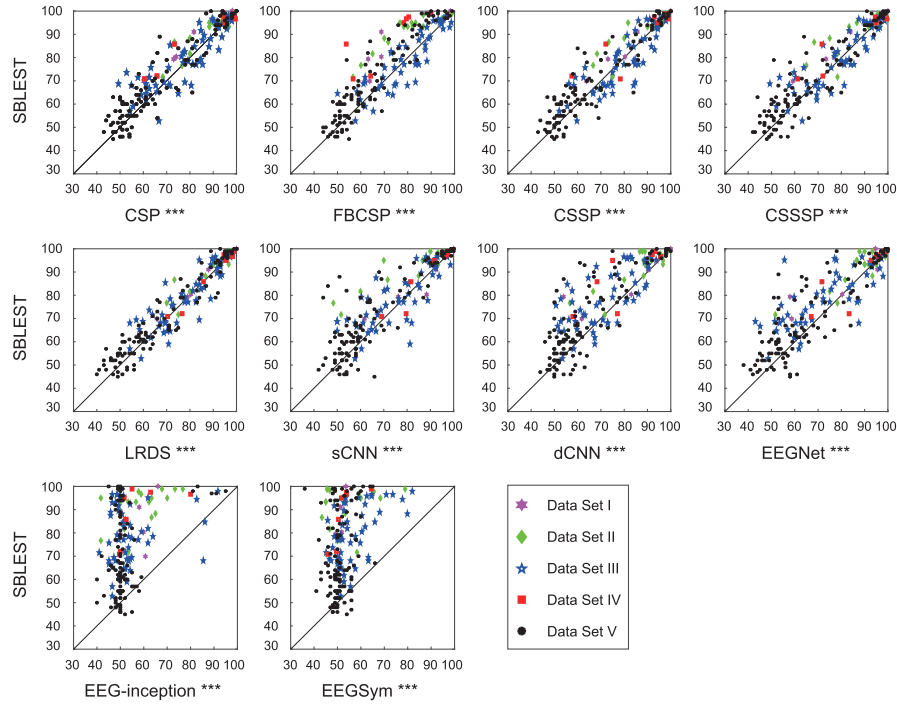


Fig. 2. Classification accuracies (%) of the competing algorithms compared to SBLEST on the 192 MI-BCI data subsets. *** $p < 0.005$. The p -values are based on the Wilcoxon signed-rank test with the Bonferroni correction.

TABLE III

MEAN CLASSIFICATION ACCURACIES (%) OF CSP, FBCSP, CSSP, CSSSP, LRDS, sCNN, dCNN, EEGNet, EEG-INCEPTION, EEGSYM AND SBLEST ON THE MI-BCI DATASETS, AND BONFERRONI-CORRECTED p -VALUES COMPARING OTHER ALGORITHMS TO SBLEST

Algorithms	Dataset I	Dataset II	Dataset III	Dataset IV	Dataset V	All MI datasets	p -value (corrected)
CSP	77.45%±13.94%	89.20%±9.56%	77.07%±13.70%	83.93%±16.80%	68.03%±17.66%	73.38%±17.20%	3.69×10^{-6}
FBCSP	72.52%±14.61%	81.23%±12.91%	78.99%±12.25%	71.83%±13.66%	65.76%±15.59%	71.33%±15.69%	1.85×10^{-12}
CSSP	80.57%±13.63%	88.40%±11.16%	79.53%±12.32%	84.64%±15.75%	68.01%±17.80%	74.09%±17.21%	5.55×10^{-3}
CSSSP	78.66%±14.95%	88.24%±10.98%	77.74%±13.58%	84.70%±15.82%	67.79%±18.18%	73.40%±17.57%	7.23×10^{-5}
LRDS	82.62%±12.56%	89.72%±9.13%	78.05%±12.43%	88.61%±11.30%	68.81%±17.97%	74.45%±17.07%	2.15×10^{-3}
sCNN	74.86%±12.20%	82.78%±15.43%	75.58%±13.86%	87.56%±10.95%	66.06%±17.29%	71.32%±17.09%	1.53×10^{-11}
dCNN	77.87%±19.43%	86.64%±11.24%	72.04%±13.25%	79.90%±14.28%	65.07%±16.30%	69.93%±16.41%	5.69×10^{-14}
EEGNet	79.03%±17.23%	86.05%±12.01%	74.31%±15.41%	86.90%±12.02%	64.94%±17.40%	70.72%±17.76%	6.04×10^{-12}
EEG-inception	58.93%±5.82%	58.70%±9.96%	53.87%±10.44%	57.38%±10.95%	51.77%±8.54%	53.40%±8.54%	7.34×10^{-30}
EEGSym	51.18%±1.69%	53.86%±9.66%	56.53%±8.63%	52.80%±5.82%	51.18%±4.86%	52.99%±6.99%	2.15×10^{-30}
sCNN-A	75.51%±16.74%	82.31%±15.64%	78.43%±14.00%	83.85%±14.48%	66.17%±18.71%	71.34%±17.17%	1.91×10^{-11}
dCNN-A	73.01%±20.21%	66.64%±18.11%	71.59%±15.27%	64.44%±17.21%	62.77%±16.10%	65.94%±16.53%	1.33×10^{-20}
EEGNet-A	75.03%±16.68%	81.67%±15.15%	73.03%±14.54%	82.64%±15.62%	66.47%±17.66%	71.00%±17.24%	6.67×10^{-14}
SBLEST	84.14%±11.61%	91.82%±7.93%	78.83%±12.19%	88.12%±12.15%	70.46%±17.51%	75.82%±16.62%	—

Methods suffixed with “-A” indicate that additional trace normalization and whitening preprocessing steps were applied to the data in this experiment.

- All the algorithms assessed in the within-subject experiment were implemented.
- The hyperparameters were determined via the grid search with stratified 5-fold cross-validation. The search space for each hyperparameter was consistent with that in the within-subject experiment (see Table I).

To reduce computational complexity, we opted for randomly selecting 50 balanced samples from the data of each subject. This resulted in a total of 2700 samples used in the cross-subject experimental analysis.

C. Results

1) *Within-Subject Decoding Performance on the MI BCI Datasets:* For all the compared algorithms, the test classification accuracies on the 192 MI BCI data subsets are shown

in Fig. 2 against that of SBLEST, where points above the diagonal indicate the superiority of SBLEST. The Wilcoxon signed-rank test shows that the SBLEST significantly outperforms all the competing algorithms (all Bonferroni-corrected p -values < 0.005).

Moreover, the mean testing classification accuracies (mean $\pm$ standard deviation) for each of the five datasets across all datasets are provided in Table III, among which the highest entries among all the algorithms are highlighted in boldface. Methods suffixed with “-A” indicate that additional trace normalization and whitening preprocessing steps were applied to the data in this experiment.

It can be seen that SBLEST achieves the highest classification performance among all the algorithms on every dataset except Dataset III and IV. The following observations are in order:

TABLE IV
MEAN CLASSIFICATION ACCURACIES (%) OF SVM, DBN, GELM, AND SBLEST ON SEED DATA SET, AND BONFERRONI-CORRECTED p -VALUES COMPARING OTHER ALGORITHMS COMPARED TO SBLEST

Algorithm	SVM	GELM	DBN	SBLEST
Mean $\pm$ std	80.10% $\pm$ 12.34%	80.06% $\pm$ 12.31%	77.83% $\pm$ 12.84%	86.97%$\pm$10.80%
p value (corrected)	2.53×10^{-6}	1.32×10^{-6}	1.99×10^{-7}	—

- 1) In terms of the mean classification accuracy across all the datasets, the SBLEST exceeds CSP, FBCSP, CSSP, CSSSP, LRDS, sCNN, dCNN, EEGNet, EEG-inception and EEGSym by 2.44%, 4.49%, 1.73%, 2.42%, 1.37%, 4.50%, 5.89%, 4.74%, 22.42%, and 22.83%, respectively.
- 2) Spatio-temporal filtering may reveal more decoding information than spatial filtering only. The mean classification accuracies of CSSP and SBLEST exceed that of CSP by 0.71% (corrected $p = 3.70 \times 10^{-3}$) and 2.44% (corrected $p = 2.23 \times 10^{-6}$), respectively. Moreover, SBLEST significantly outperforms LRDS (corrected $p = 2.15 \times 10^{-3}$). These favorable results for spatio-temporal filtering compared to spatial filtering only suggest that optimizing individual-specific temporal filters is important for decoding.
- 3) Channel-specific temporal filters may be preferable to channel-invariant temporal filters. This is evidenced by the higher mean classification accuracies of CSSP and SBLEST over that of CSSSP by 0.69% (corrected $p = 7.90 \times 10^{-3}$) and 2.42% (corrected $p = 4.37 \times 10^{-5}$), respectively. These results are in line with the prior findings that the specific frequency bands comprising information for decoding movement intentions may vary across channels [42].
- 4) Simultaneous optimization of the spatio-temporal filters and the classifier may be more advantageous to the conventional two-step decoding approaches. This is manifested by the superior mean classification accuracies of SBLEST over those of CSP and CSSP by 2.44% (corrected $p = 2.23 \times 10^{-6}$) and 1.73% (corrected $p = 4.37 \times 10^{-5}$).
- 5) Deep learning algorithms including sCNN, dCNN, and EEGNet do not lead to better classification performance than other decoding methods. Among the considered deep learning algorithms, sCNN achieves the highest mean classification accuracy. However, the performance is consistently poorer than non-deep learning algorithms. For EEG-inception and EEGSym, which have deeper and more complex network structures, the binary classification results are close to random.
- 6) The performance of dCNN with trace normalization and whitening (dCNN-A) was significantly worse than that without ($p = 9.19 \times 10^{-4}$). In contrast, sCNN and EEGNet maintained similar performance with and without these preprocessing steps.

Furthermore, Tables S2 and S3 present the results of F1 scores and ITRs, where SBLEST still outperforms other competitive algorithms, providing additional evidence of the algorithm's superiority.

2) *Decoding Performance on the SEED Dataset:* For the SEED dataset, the classification accuracies of all the

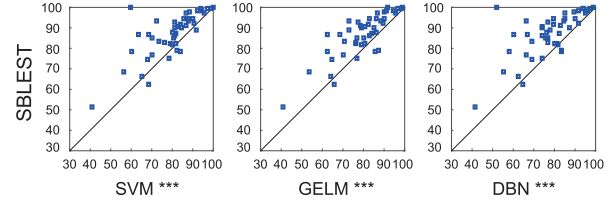


Fig. 3. Classification accuracies (%) of the competing algorithms compared to SBLEST on the 45 data subsets from the SEED dataset. *** $p < 0.005$. The p -values are based on the Wilcoxon signed-rank test with the Bonferroni correction.

compared algorithms against SBLEST on all the 45 data subsets are depicted in Fig. 3. The mean classification accuracies are provided in Table IV. On this emotion recognition dataset, SBLEST substantially outperforms the compared algorithms; the improvement in the mean classification accuracy is 6.87% (corrected $p = 2.53 \times 10^{-6}$) over the best of the compared algorithms. Again, DBN, which is a deep learning algorithm, performs worse than non-deep learning algorithms.

3) *Cross-Subject Decoding Performance on the OpenBMI Dataset:* Table V presents the cross-subject classification accuracies of all algorithms evaluated on the OpenBMI dataset. Consistent with the within-subject results, SBLEST outperforms all other algorithms. However, in contrast to the within-subject analysis, sCNN, dCNN, and EEGNet demonstrate superior performance when compared to CSP. Moreover, the performance of EEGSym is significantly inferior to other methods. Similar conclusions can be drawn from the results of F1 scores and ITRs, as presented in Table S4.

4) *Visualization of Spatio-Temporal Filters:* The optimal spatio-temporal filters from SBLEST can be readily visualized and their validity can be assessed against prior neurophysiological knowledge. Below we illustrate this by visualizing the spatio-temporal filters for different algorithms across various datasets and experiments compared to SBLEST. In each case, the scalp maps and frequency response curves correspond to the latent signals that show the highest positive and negative correlation, respectively, with the class labels on the training set. All figures are included in Supplemental Material, available online.

For within-subject MI analysis, Fig. S3 shows the scalp maps of the spatial filters and the frequency response curves of the temporal filters optimized by each algorithm for three representative subjects, i.e., *ay*, *S3*, and *SUB93*. For each subject, the classification accuracies associated with different algorithms are provided in Table S5.

Across all the subjects of within-subject motor imagery analysis depicted in Fig. S3, the scalp maps obtained by SBLEST are predominantly centered on areas well aligned with the

TABLE V

CROSS-SUBJECT ACCURACIES (%) OF CSP, FBCSP, CSSP, CSSSP, LRDS, sCNN, dCNN, EEGNet, EEG-INCEPTION, EEGSYM, AND SBLEST ON OPENBMI DATASET

Algorithms	CSP	FBCSP	CSSP	CSSSP	LRDS	sCNN	dCNN	EEGNet	EEG-inception	EEGsym	SBLEST
Accuracies	63.44%	67.52%	66.78%	65.85%	66.52%	64.93%	66.74%	67.26%	61.42%	54.37%	69.41%

TABLE VI

MEAN CLASSIFICATION ACCURACIES (%) AND p -VALUES BASED ON THE WILCOXON SIGNED-RANK TEST COMPARING SBLES TO SBLEST

Algorithms	Dataset I	Dataset II	Dataset III	Dataset IV	Dataset V	Dataset VI	All datasets	p -value
SBLES	84.34%±12.29%	89.51%±10.06%	77.03%±12.92%	87.04%±12.89%	68.86%±17.56%	86.63%±11.09%	76.53%±16.68%	2.94×10^{-6}
SBLEST	84.14%±11.61%	91.82%±7.93%	78.83%±12.19%	88.12%±12.15%	70.46%±17.51%	86.97%±10.80%	77.94%±16.27%	—

corresponding task-related areas in the motor cortex according to neurophysiology. In particular, while all the algorithms reveal task-related areas in the scalp maps for subject *ay*, LRDS and SBLEST exhibit spatial patterns with a higher signal-to-noise ratio and more focused regions of interest, which in turn lead to the superior classification accuracies compared to the other algorithms, with SBLEST's being the highest. It can also be observed that the frequency response curves of the associated temporal filters optimized by SBLEST are not flat for subjects *S3* and *SUB93*. These subject-specific temporal filters, in conjunction with the spatial filters, translate to the enhancement of the classification performance of SBLEST over other algorithms.

Furthermore, to gain insight into whether individualized spatial filters from SBLEST could still produce meaningful global spatial patterns on a group level in the within-subject analysis, we computed the average spatial filters for each binary classification task over all participants. The resulting mean spatial filters are visualized in Fig. S4, which demonstrates that the averaged scalp maps obtained from SBLEST remain predominant in the relevant task-related regions of the motor cortex.

For emotion recognition, Fig. S5 presents the scalp maps of the spatial filters and the frequency response curves of the temporal filters from SBLEST. As can be seen, the prominent areas in the scalp maps for the positive emotion are the lateral temporal areas. By contrast, the scalp maps for the negative emotion show higher weights in the occipital, parietal, and pre-frontal areas. These patterns are in line with those reported in the prior EEG-based emotion recognition studies [3]. Nonetheless, there are also considerable variations in the EEG spatio-temporal filters across subjects, despite the identical film clips that they were watching. This suggests that optimizing subject-specific spatio-temporal filters is essential for achieving superior decoding performance.

Fig. S6 shows the scalp maps of the spatial filters and the frequency response curves of the temporal filters in the cross-subject analysis. In terms of the optimal temporal filters, the frequency responses are highly consistent across CSSP, CSSSP, and SBLEST. However, SBLEST stands out for spatial filters, as it produces more centralized patterns than CSSP and fewer dipolar patterns than CSSSP, leading to its superior classification performance.

5) *Spatio-Temporal Filtering Versus Spatial Filtering*: In order to more directly assess the benefit of optimizing temporal filters in SBLEST, we additionally run prediction analysis by fixing the order of the temporal filters in SBLEST to zero,

i.e., $\tau = 0$. The resulting algorithm, where only the spatial filters and the classifier are optimized, is termed SBLES. For SBLES, all the processing steps are the same as SBLEST, except that the empirical augmented covariance matrices are replaced by the empirical covariance matrices. The mean testing classification accuracies for SBLES are shown in Table VI in comparison to SBLEST. This result shows that SBLEST achieves significantly better prediction performance than SBLES, supporting the need of optimizing temporal filters for EEG decoding.

6) *Sparsity Induced by SBLEST*: We next seek to verify whether SBLEST is able to induce sparsity in $\mathbf{W}$ as desired. As a metric to quantify the sparsity strength achieved by SBLEST irrespective of the number of EEG channels, we define the relative cardinality as L/KC . The relative cardinalities on the five MI BCI datasets are shown in Fig. 4(a). Across all six datasets, the mean relative cardinalities are lower than 21%, demonstrating SBLEST yields substantial sparsity in terms of the number of the latent signals.

7) *Relationship Between SBL-Optimized λ and Prediction Performance*: The noise-related hyperparameter λ is optimized via SBL in SBLEST. To verify that this optimization leads to good prediction performance on the test set, we fix λ to each value in $\{10^{-5}, 10^{-4.75}, 10^{-4.5}, \dots, 1\}$, use the update rules in SBLEST to optimize the other unknown parameters and hyperparameters, and then assess the prediction performance on the test set. It is expected that the prediction performance of SBLEST should be close to the best prediction performance among the considered λ 's. Fig. 4(b) shows the classification accuracy as λ varies for Dataset I. It can be seen that for each subject in this dataset, SBLEST-optimized λ can indeed achieve good out-of-sample prediction performance.

8) *Computational Cost*: In SBLEST, the computational cost is mainly associated with the high-dimensional matrix multiplication and inversion operations determined by the number of the trials M , the number of EEG channels C , and the order of the temporal filters K . More specifically, let I denotes the number of iterations for SBLEST to converge, it can be shown that the total computational complexity is $O(\max(IMK^3C^3, IM^2K^2C^2, IM^3))$, which becomes IMK^3C^3 when C and K are large compared to M .

For comparison, Table VII provides the runtimes of SBLEST ($\tau = 2, K = 2$) versus neural network based algorithms on different datasets. EEGNet is coded in Tensorflow using Keras API, and sCNN and dCNN are coded in Pytorch. All algorithms are

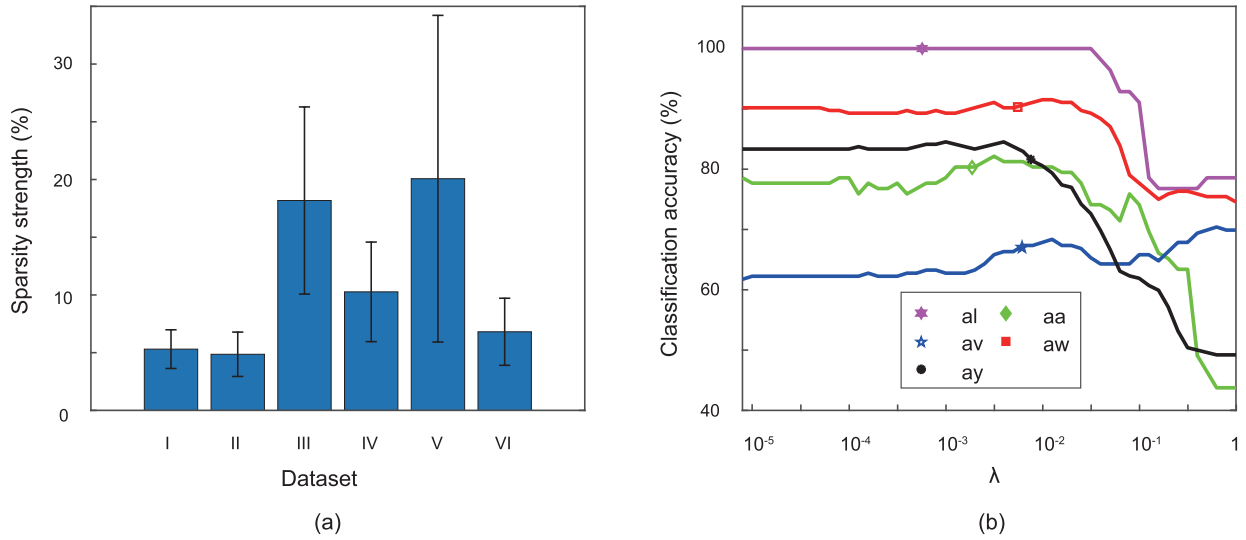


Fig. 4. (a) Sparsity strength quantitated by the relative cardinality $L/(KC)$ (%) on the six MI BCI datasets (mean $\pm$ standard deviation). (b) Classification accuracies for the five subjects on the test sets in Dataset I with varying λ . The optimal λ determined by SBLEST based on the training set is indicated on the curve for each subject.

TABLE VII

RUNTIMES (SEC) OF SBLEST, sCNN, dCNN, AND EEGNet APPLIED TO MI DATASETS WITH VARYING CHANNEL NUMBER C AND TRAINING SAMPLE SIZE M

subject	al	aa	av	aw	ay	k3	k6 and L1	A01 – A09	S1 – S7	SUB1 – SUB108
# channels C	49	49	49	49	49	60	60	22	32	20
# samples M	224	168	84	56	28	90	60	144	60	100
SBLEST	152.41 s	337.53 s	136.45 s	154.80 s	111.70 s	297.55 $\pm$ 2.49 s	244.20 $\pm$ 6.10 s	11.72 $\pm$ 5.64 s	29.11 $\pm$ 16.61 s	8.90 $\pm$ 7.09 s
sCNN	53.36 s	27.97 s	28.90 s	16.38 s	18.92 s	45.75 $\pm$ 27.65 s	22.71 $\pm$ 8.12 s	43.12 $\pm$ 18.92 s	24.88 $\pm$ 2.94 s	112.10 $\pm$ 61.34 s
dCNN	40.25 s	120.75 s	19.93 s	18.49 s	13.44 s	22.55 $\pm$ 6.15 s	17.04 $\pm$ 3.58 s	39.06 $\pm$ 16.55 s	20.86 $\pm$ 5.19 s	109.24 $\pm$ 36.42 s
EEGNet	190.35 s	144.88 s	75.85 s	52.68 s	30.03 s	384.56 $\pm$ 0.80 s	258.62 $\pm$ 4.35 s	177.45 $\pm$ 1.73 s	128.64 $\pm$ 0.91 s	142.11 $\pm$ 25.45 s

run on a Windows PC with a 3.70-GHz Inter Core (TM) i7-8700 K CPU and 16-GB RAM, in MATLAB®2018b. It can be observed from the table that while SBLEST is computationally efficient when the number of channels is relatively low (i.e., 20, 22, and 32), there is a significant rise in runtime when the channel number is increased to 49 and 60 in Datasets I and II, respectively.

IV. DISCUSSIONS

A. Spatio-Temporal Filtering is Important for EEG Decoding but Performance Varies Across Methods

Among the algorithms considered in this work, CSP and LRDS only seek to optimize spatial filters while assuming the optimal frequency band is known *a priori*, which is hardly the case. Moreover, although CSP is an effective spatial filtering algorithm, its optimization criterion, i.e., variance ratio maximization, is only loosely connected to the decoding objective. By contrast, LRDS learns spatial filters to minimize the decoding error, albeit with the sparsity measured by the nuclear norm. Consequently, LRDS is expected to outperform CSP from a decoding standpoint, which is supported by the average classification accuracies in Table III and cross-subject classification accuracies in Table V.

In addition to spatial filtering, FBCSP, CSSP, CSSSP, sCNN, dCNN, EEGNet, and SBLEST incorporate temporal filtering

in an attempt to further improve the decoding performance. In particular, FBCSP, CSSP, and CSSSP extend CSP to consider temporal filter optimization and thereby inherit CSP's drawbacks for spatial filter optimization. Alternatively, sCNN, dCNN, and EEGNet take a neural network approach where spatio-temporal filters are optimized as a part of an end-to-end decoding framework. Despite the conceptual appeal of these neural network-based approaches, the results presented in Table III suggest that they do not necessarily yield better within-subject decoding performance than the CSP-based algorithms, or even more surprisingly, the vanilla CSP as opposed to the results reported in some earlier studies where algorithms were tested on a smaller number of datasets. Our results therefore suggest that these deep learning methods may be less capable of learning subject-specific spatio-temporal filters than shared spatio-temporal filters across subjects.

This striking finding highlights shortcomings of the existing neural network approaches. First, the numbers of weight parameters that need to be learned in sCNN, dCNN, and EEGNet are exceedingly larger than other methods. This added degree of freedom may pose difficulty for the models to generalize to unseen data. Second, gradient based algorithms are employed for parameter optimization in the neural network based approaches, which may converge to stationary points with poor decoding performance compared to the global optimal solutions. Although recent analyses suggest that stochastic gradient descent for

overparameterized deep learning models may be able to achieve excellent generalization performance in certain scenarios [43], it remains to establish whether this property holds for CNN, dCNN and EEGNet and our within-subject classification results certainly indicate otherwise with the datasets considered in our study.

Analogous to the neural network based approaches, SBLEST also optimizes the spatio-temporal filters in an end-to-end manner to minimize the decoding error, hence flexible. Although SBLEST employs a non-convex algorithm for parameter estimation, it has been shown that under mild conditions the cost function has a unique stationary point located at the global optimum [28]. Intuitively, the local minima are smoothed away via the marginalization step in Equation (23). Compared to the convex nuclear norm regularization as in LDRS, SBLEST provides a better surrogate of the rank penalty, which possesses the identical global optimum as the rank function [28]. Our results indicate that SBLEST significantly outperforms all other compared algorithms in both the within-subject and cross-subject classification accuracies.

B. Sparse Bayesian Learning Provides a Powerful Framework for Decoding Rhythmic Brain Activities

SBL provides a powerful framework for learning parsimonious linear latent variable models from EEG, with applications encompassing electromagnetic source imaging [44], [45], probabilistic generative modeling of EEG (e.g., oscillations and ERPs) [46], [47], and EEG decoding [9], [48]. In essence, SBL is an empirical Bayes paradigm that imposes parameterized prior on the latent variables and enforces sparsity via maximizing the marginal likelihood (also known as Type-II maximum likelihood or evidence maximization). Notably, marginal likelihood maximization has been applied with enormous success for hyperparameter learning [49]. Moreover, its deep connection with cross-validation was recently elucidated [50], which is also supported by our result in Fig. 4.

It should be noted that existing SBL approaches for EEG decoding are predominantly designed for linear models with vector-valued inputs, e.g., snapshots of EEG amplitudes [51], [52]. While these approaches may be suited for ERP analysis, they are unable to deal with volume conduction for rhythmic brain activities, which entails a nonlinear model with respect to the spatial filters when the spectral power is considered as the feature [53]. However, for EEG decoding based on spatio-temporal filtering, we have shown in Section II-C that the problem can be reframed as rank-penalized linear matrix regression that works on matrix-valued data, i.e., the augmented covariance matrix. As such, further development of the SBL technique is needed to handle this matrix-valued data. In this paper, we developed an EM-like algorithm based on our prior work to solve the rank-penalized matrix regression problem. The excellent results on decoding imagined movements and emotions demonstrate that its favorable theoretical properties are able to translate to improved performance in practical settings. This also implies that the EEG data may be well described by

a low-rank structure and recovering this low-rank structure is advantageous to EEG decoding.

C. Decoding Performance on Small-Sample Datasets

SBL has an advantage of being able to efficiently utilize data compared to methods that require cross-validation for hyperparameter estimation. As a result, it is expected that SBLEST can outperform other methods on datasets with limited sample sizes. To test this hypothesis, we conducted additional experiments to compare SBLEST with other algorithms. Specifically, we selected data subsets where the sample size (M) was small relative to the number of elements in the spatial covariance matrix (C^2), using the criterion of $M/C^2 < 0.06$, and evaluated the classification performance of each algorithm on these subsets ($N = 21$). The results are presented in Table S6, where the classification performance of SBLEST on these subsets was significantly better than other algorithms ($p < 0.05$), except for LRDS. These results suggest that SBLEST may exhibit superior performance on small datasets, although further comparisons on more datasets are warranted to validate this finding.

D. Limitations and Future Work

The computational cost of SBLEST rises significantly when the channel number is large. Developing more scalable algorithms to solve the affine constraint low-rank minimization problem is therefore a natural next step for our future studies.

In SBLEST, we assume $\gamma = 1$. However, if γ are treated as unknown hyperparameters, in principle an updated rule for γ can be similarly derived under the evidence maximization framework. This can be leveraged to sparsify elements of the spatial or temporal filters by driving the corresponding γ_j 's to zero, which may further enhance the generalization performance. Sparse spatio-temporal filters may also be useful in practical applications for interpretation purpose. Moreover, despite the superiority of SBLEST over state-of-the-art EEG decoding algorithms in our experiments, the order of the temporal filters considered in this work is low for computational reasons. Higher-order temporal filters will be investigated in our future studies.

In terms of cross-subject decoding, SBLEST demonstrated superior performance on the OpenBMI dataset with 54 subjects, outperforming all other algorithms. However, it remains unclear whether this superiority remains consistent as the sample size increases. To address this uncertainty, we conducted an additional cross-subject analysis using the Physionet MI dataset [54], which comprised 105 subjects and 4725 trials (refer to Supplemental Material, available online for dataset details and analysis procedure). Table S7 presents the cross-subject classification accuracy results for the left versus right fist motor imagery task across all methods. While SBLEST outperformed all other non-deep learning methods, deep learning techniques including sCNN, dCNN, and EEGNet achieved slightly higher accuracies than SBLEST. This finding suggests that deep learning methods may derive greater benefits from larger sample sizes in the context of cross-subject decoding. However, it is important

to conduct further analyses using additional datasets to draw definitive conclusions.

Our study focuses on evaluating the performance of a linear regression model for decoding EEG signals in binary classification problems. Nonetheless, logistic regression models may be more suited for handling classification problems and may offer improved performance. In our future work, we plan to develop a logistic regression counterpart of SBLEST, based on its Type-I formulation in Equation (31). Moreover, we plan to address multi-class classification problems by using strategies such as one-versus-others, one-versus-one, or ensemble techniques.

V. CONCLUSION

In this paper, we introduced SBLEST, an end-to-end algorithm for EEG decoding. SBLEST optimizes spatio-temporal filters and the classifier simultaneously within a principled sparse Bayesian learning framework to maximize prediction accuracy.

To evaluate SBLEST's effectiveness, we conducted a systematic analysis on five motor imagery EEG datasets and an emotion recognition EEG dataset. Our results demonstrate that SBLEST significantly outperformed several state-of-the-art EEG decoding algorithms. Additionally, the learned spatio-temporal patterns were amenable to visualization and interpretation, enabling the extraction of neurobiological or potentially clinical insight. Taken together, our findings demonstrate the potential of SBLEST for decoding EEG signals in BCIs and biomarker identification for brain disorders.

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